

軟體測試

期末考

考試中，請勿參閱資料。

1. Please explain Beizer testing level 0, 1, 2, 3, and 4.

Level 0: (2pts/2)

Level 1: (2pts/4)

Level 2: (2pts/6)

Level 3:(2pts/8)

Level 4:(2pts/10)

2. Below is a graph, defined by the sets of nodes, initial nodes, final nodes, edges, and defs and uses.

Following the graph, there also is a collection of test paths. Answer the following questions about the graph.

Graph: $N=\{0,1,2,3,4,5,6,7\}$

$N_0=\{0\}$

$N_f=\{7\}$

$E=\{(0,1),(1,2),(2,7),(2,3),(2,4),(3,4),(4,5),(5,6),(4,6),(6,2)\}$

$\text{def}(0)=\text{def}(5)=\text{use}(4)=\text{use}(2)=\{x\}$

Test paths:

$t_0=[0,1,2,7]$

$t_1=[0,1,2,4,6,2,7]$

$t_2=[0,1,2,4,5,6,2,7]$

$t_3=[0,1,2,3,4,6,2,7]$

$t_4=[0,1,2,3,4,5,6,2,7]$

$t_5=[0,1,2,3,4,5,6,2,4,5,6,2,7]$

(a) Draw the graph. (5pts/15)

- (b) List all of the du-paths with respect to x. (Note: Include all du-paths, even those that are subpaths of some other du-path). (5pts/20)
- (c) For each test path, determine which du-paths that test path tours. For this part of the exercise, you should consider both direct touring and sidetrips. Hint: A table is a convenient format for describing this relationship. (5pts/25)
- (d) List a minimal test set that satisfies all-defs coverage with respect to x. (Direct tours only.) Use the given test paths. (5pts/30)

(e) List a minimal test set that satisfies all-uses coverage with respect to x . (Direct tours only.) Use the given test paths. (6pts/35)

(f) List a minimal test set that satisfies all-du-paths coverage with respect to x . (Direct tours only.) Use the given test paths. (5pts/40)

3. Use the following predicate to answer the following questions.

$$p = \neg(a \vee (\neg a \wedge b) \vee (\neg b \wedge \neg c))$$

(a) Identify the clauses that determine predicate p . (5pts/45)

(b) Compute (and simplify) the conditions under which each of the clauses determines predicate p . (5pts/50)

(c) Write the complete truth table for all clauses. Label your rows starting from 1. Use the format that row 1 should be all clauses true. You should include columns for the conditions under which each clause determines the predicate, and also a column for the predicate itself. (5pts/55)

- (d) Identify all pairs of rows from your table that satisfy *general active clause coverage* (GACC) with respect to each clause. (5pts/60)
- (e) Identify all pairs of rows from your table that satisfy *correlated active clause coverage* (CACC) with respect to each clause. (5pts/65)
- (f) Identify all pairs of rows from your table that satisfy *restricted active clause coverage* (RACC) with respect to each clause. (5pts/70)

- (g) Identify all 3-tuples of rows from your table that satisfy *general inactive clause coverage (GICC)* with respect to each clause. Identify any infeasible GICC test requirements. (5pts/75)
- (h) Identify all 3-tuples of rows from your table that satisfy *restricted inactive clause coverage (RICC)* with respect to each clause. Identify any infeasible RICC test requirements. (5pts/80)

4. We have a program that accepts three integer input parameters a , b , and c . The program then solves $ax^2+bx+c=0$ and returns the solutions if any. Please show how to apply the input space partitioning technique to design test input for completeness and disjointness among blocks. (10pts/90)

5. Assume that we have the following program with two mutants.

Original program:

```
public static int count(int[] x, int y) {  
    int c = 0;  
    for (int j = 0; j < x.length; j++)  
        if (x[j] == y) c++;  
    return c;  
}
```

(a) Please write down a test input that kills mutant 1 in the following. Explain why it kills the mutant? (5pts/95)

Mutant 1:

```
public static int count(int[] x, int y) {  
    for (int j = 0; j < x.length; j++)  
        if (x[j] == y) c++;  
    return c;  
}
```

(b) Please write down a test input that kills mutant 2 in the following! Please explain why it kills the mutant. (5pts/100)

Mutant 2:

```
public static int count(int[] x, int y) {  
    int c = 0;  
    for (int j = 0; j <= x.length; j++)  
        if (x[j] == y) c++;  
    return c;  
}
```